Aptitude ends in 27 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 27 min

Go to Analog ›

Digital

‹ Previ

20. Rob and his old calendars

Rob is looking through some old calendars and finds that the **21st of August 1995** was a Monday.

If 1996 was a leap-year, what day of the week was the **24th of August 1998**?

Answer Options

Select any one option

- Monday
- Thursday
- Friday
- Sunday

Aptitude ends in 30 min
Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Analog
Starts in 30 min
Go to Analog >
Digital

2. A herbal energy drink

Argument: A specific brand's herbal energy drink was given to a group of 50 consumers and a study was conducted to prove the benefits of the drink on the mood of a person. Consumers were put in time bound situations that and then given a glass of the drink to help them focus. The test results were out and 80 percent of the consumers agreed the energy drink in fact did help with their productivity. Hence, it can be proven that the specific brand's energy drink on which the survey was conducted is the best energy drink in the market.

The above given argument's reasoning is most vulnerable to criticism on the grounds that it fails to consider which of the given choices?

Answer Options

Clear Answer

Select any one option

- Any of the consumers that participated in the survey are allergic to some herbs in the drink.
- The herbs in the energy drink is not very harmful to the health of the consumer.
- The price of the energy drink when compared to other brands is high.
- Customers had the opportunity to taste the energy drink from various other brands as well

‹ Previous Next ›

1. Drip irrigation for lawns

Gardener: When adding lawns for the new house, we should install drip irrigation system in addition to the sprinklers. This will ensure that the lawns don't dry out in case someone forgets to water the lawn and reduce damage in the garden. The cost of installing drip irrigation system is relatively lower when compared to redoing the lawns due to the damage caused by under watering.

House owner: Installing drip irrigation systems are very expensive and unnecessary and seem like a wasteful expense.

Which of these choices, if true, could best be used to support the Gardener's proposal in the face of the House owner's objection?

Answer Options

Select any one option

Clear Answer

- The use of sprinklers can still reduce the longevity of the lawns due to underwatering.
- Severe rains can cause the lawns to rot
- During a snow storms, all the lawn gets slightly damaged.
- Drip irrigation system can save the lawns when nobody is there to water it.

3. Jia's pocket money

Jia receives an amount **X** as pocket money every month. Of the total amount received, Jia spends 20% on necessary purchases and 5% of the remaining amount on transportation.

If she is left with Rs. 1520, the what is **X** in the given context?

Answer Options

Select any one option

Clear Answer

- Rs. 2000
- Rs. 5500
- Rs. 2500
- Rs. 6080

Aptitude ends in 29 min

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Analog

Starts in 29 min

Go to Analog ›

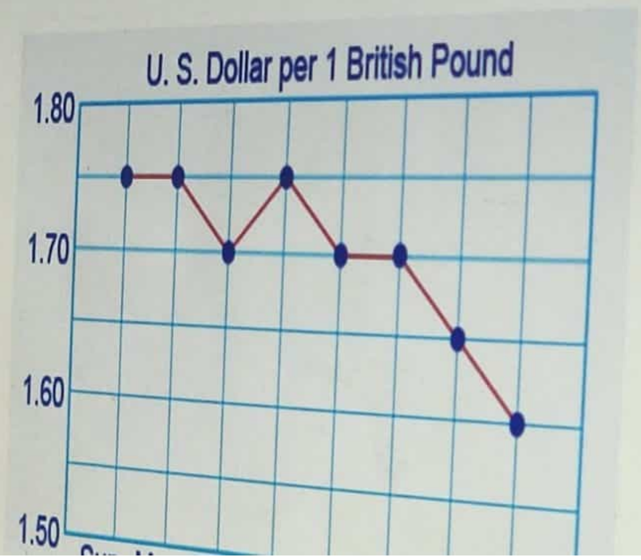
Digital

5. Profits made by Mrs. Lind...

The given line graphs show the exchange rate dynamics during one week.

On Tuesday, Mrs. Lindsay bought 200 British Pounds using US Dollars. On Wednesday, she exchanged these 200 British Pounds back into US Dollars. What profit did Mrs. Lindsay make?

Note: Assume that there was zero transaction cost.



Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 29 min

Go to Analog ›

Digital

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4. A hexagonal table

Six friends Adam, Abe, Pepa, Jill, Bob and Ron are sitting around a hexagonal table each at one corner facing the centre. Adam is second to the left of Ron. Abe is neighbour of Pepa and Jill. Bob is second to the left of Jill.

Who is sitting to the opposite of Adam?

Answer Options

Clear Answer

Select any one option

- Bob
- Jill
- Ron
- Pepa

Aptitude ends in 30 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 30 min

Go to Analog ›

Digital

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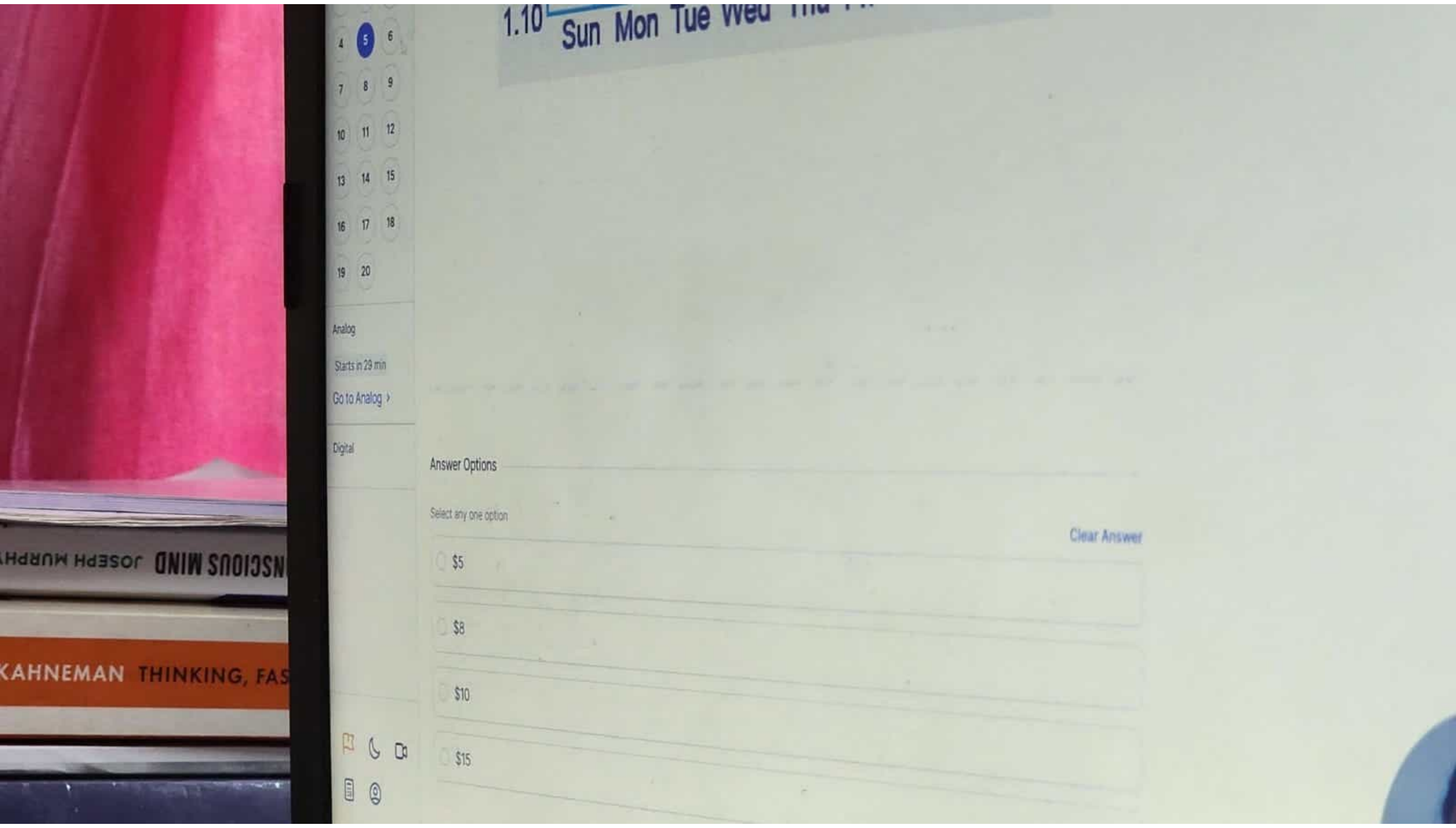
If she is left with Rs. 1520, the what is **X** in the given context?

Answer Options

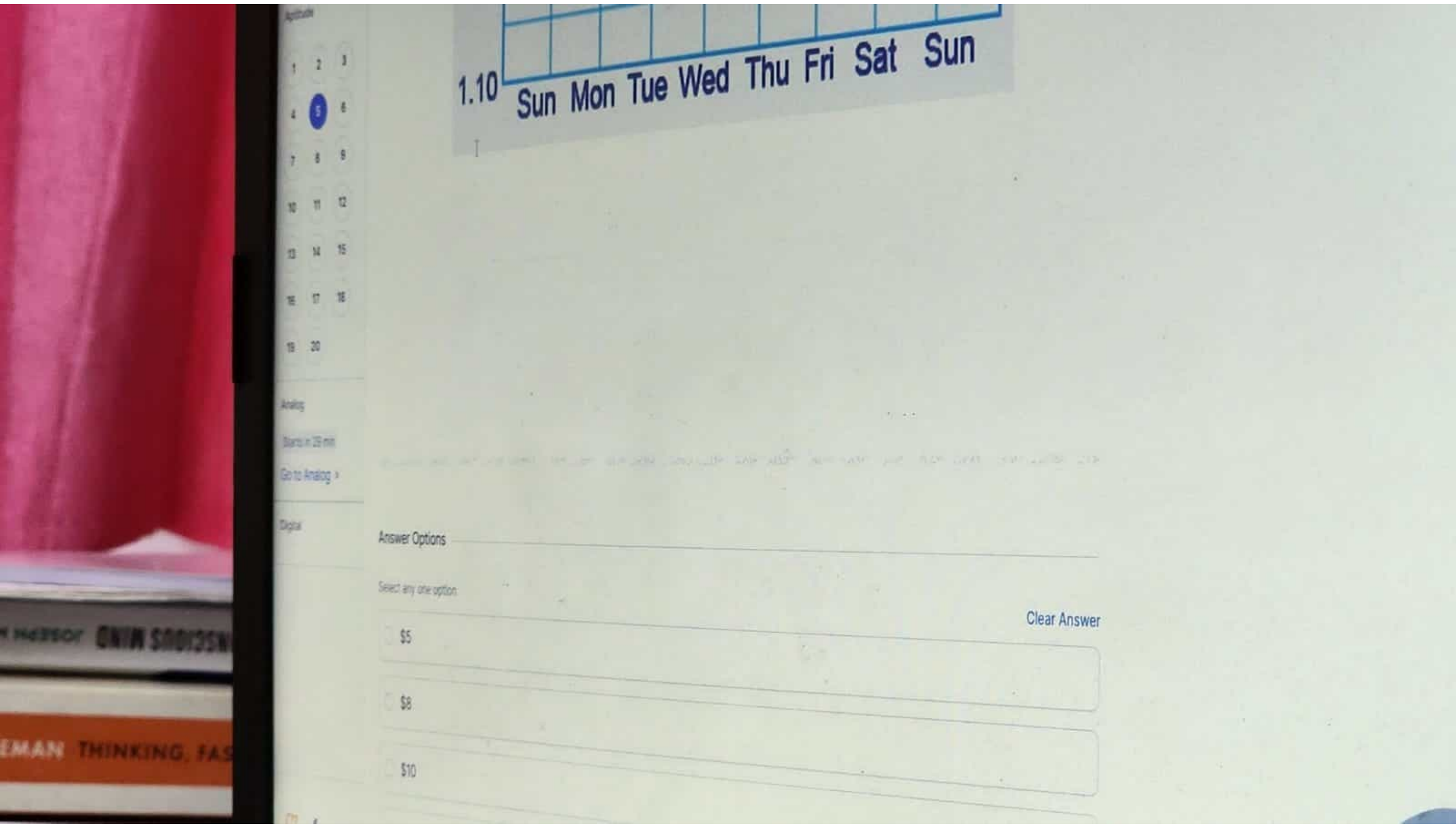
Clear Answer

Select any one option

- Rs. 2000
- Rs. 5500
- Rs. 2500
- Rs. 6080



1.10
Sun Mon Tue Wed
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Analog
Starts in 29 min
Go to Analog ›
Digital
Answer Options
Select any one option
Clear Answer
$5
$8
$10
$15
KAHNEMAN


1.10
Sun Mon Tue Wed Thu Fri Sat Sun
Analog
Go to Analog
Digital
Answer Options
Select any one option
Clear Answer
$5
$8
$10


Aptitude
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Analog
Starts in 29 min
Go to Analog ›
Digital
Euros per 1 British Pound
1.40
1.30
1.20
1.10
Sun Mon Tue Wed Thu Fri Sat Sun

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 29 min

Go to Analog ›

‹ Previous

6. A mocktail

Bob is preparing a mocktail for his friends by mixing 10 parts water and 5 parts of a mocktail mix in a large jug.

How much of the mixture should he draw out of the jug and replace it with the mocktail mix to ensure that the mixture if half water and half mocktail mix?

Note: *Assume that the jar contains 15 litres of the mixture.*

Answer Options

Clear Answer

Select any one option

- 1/4
- 1/8
- 1/2

Aptitude ends in 29 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 29 min

Go to Analog ›

Digital

6. A mocktail

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How much of the mixture should he draw out of the jug and replace it with the mocktail mix to ensure that the mixture if half water and half mocktail mix?

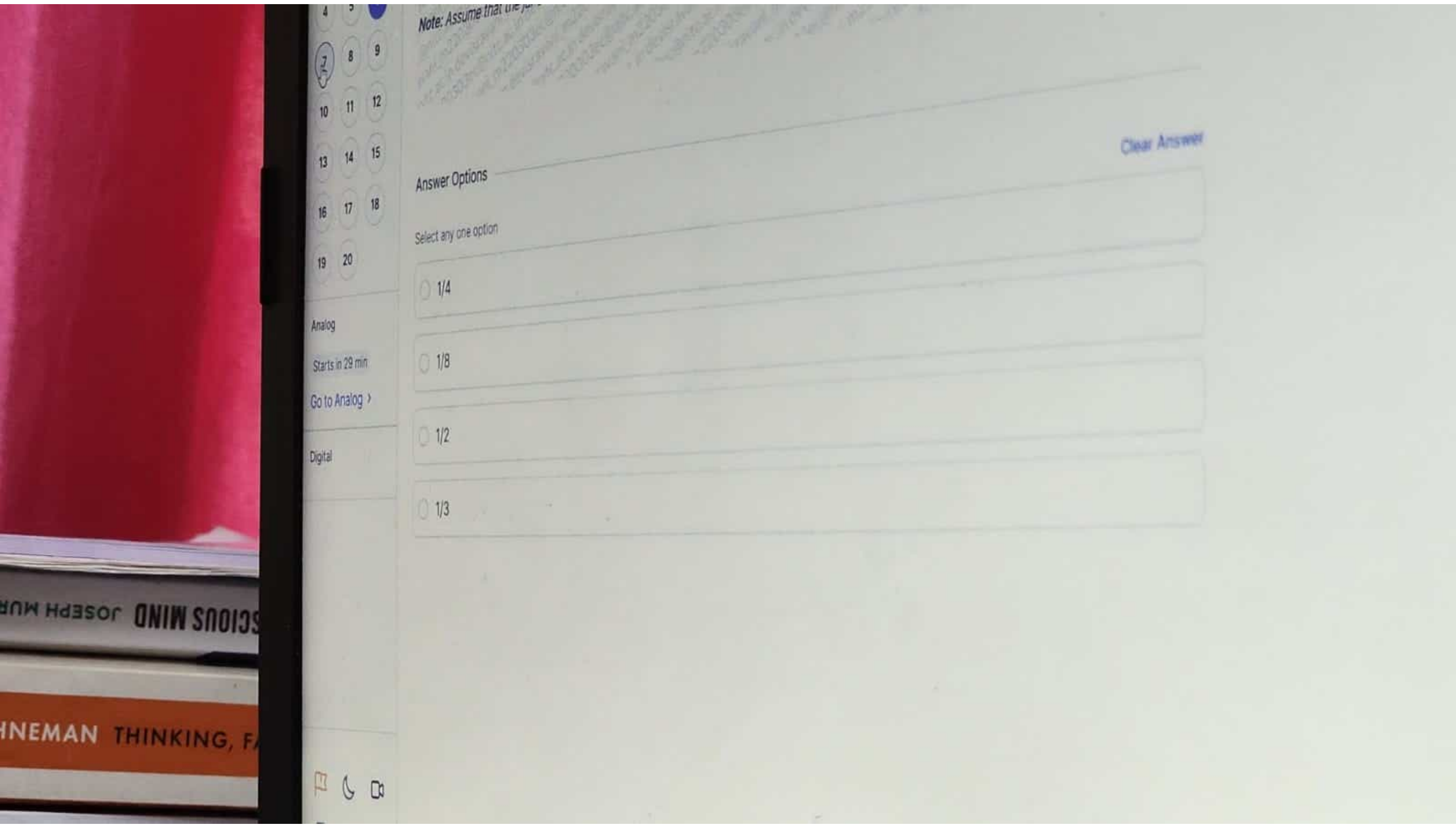
Note: *Assume that the jar contains 15 litres of the mixture.*

Answer Options

Clear Answer

Select any one option

- 1/4
- 1/8
- 1/2
- 1/3


8 9
10 11 12
13 14 15
16 17 18
19 20
Analog
Starts in 29 min
Go to Analog ›
Digital
Answer Options
Clear Answer
Select any one option
1/4
1/8
1/2
1/3

Aptitude ends in 29 min

8. A rectangular plot

A rectangular plot has been alloted by the government for the construction of a embassy building. The length of the plot is increased by 10% to equip a cafeteria.

By what percent should the breadth of the plot be decreased such that the area of the plot is same as before?

Answer Options

Select any one option

Clear Answer

- 120/13
- 100/11
- 120/11
- 100/9

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 29 min

Go to Analog >

Digital

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Next >

Submit test

Aptitude ends in 29 min

Aptitude

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Analog

Starts in 29 min

Go to Analog ›

Digital

Submit test

‹ Previous Next ›

7. Bob's shop

Bob is a retailer selling rice in his shop. He mixes two varieties of rice R1 and R2 together in the ratio 4:7 and sells this mixture at a price of Rs. 17 per kg. Due to this, he incurs a loss of 15%.

In the given context, what is the cost price per kg of R1 if the cost price of R2 is Rs. 22 per kg?

Answer Options

Clear Answer

Select any one option

- Rs. 15
- Rs. 16.5
- Rs. 17
- Rs. 17.5

Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 28 min

Go to Analog ›

Digital

Submit test

9. Five students

Five students are made to stand in a row as given in the following statements.

Statements:

1. A and B are standing next to each other.
2. C and D are standing next to each other.
3. A is not standing next to E who is standing at the left end of the line.
4. B is standing in the second position from the right.
5. C is to the right of both D and E.
6. B and C are standing together.

Which of the given choices is/are **TRUE** regarding the following conclusions made from the given statements?

Conclusions:

1. B is placed between C and A.
2. A is placed between C and B

Answer Options

Select any one option

- Only (1) conclusion is incorrect
- Only (2) conclusion is incorrect
- Either (1) or (2) is incorrect
- Neither (1) nor (2) is incorrect

Previous Next

11. Grids A and B

The two grids **A** and **B** given below follow a certain pattern. Analyse **A** and determine the correct value of '?' in **B**.

3	9	4
8	2	14
3	12	6

A

4	13	5
?	4	15
2	12	6

B

Answer Options

Select any one option

10

11

12

14

Clear Answer

10. Bella arranging books

Bella is arranging her books on a new bookshelf. She has placed B1 in the 24th position from the left and B2 in the 27th position from the right.

What is the **minimum number of books** that Bella owns if only B3 is exactly in between B1 and B2?

***Note:** Bella is placing all her books on the bookshelf.*

Answer Options

Select any one option

- 58
- 55
- 48
- 50

‹ Previous Next ›

12. Cats and dogs

Given below are 4 statements followed by 4 conclusions numbered 1 to 4. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Choose the most appropriate option for the given conclusions that logically follows from the given statements, disregarding commonly known facts.

Statements:

Some cats are dogs. No dog is a frog. All frogs are bats. Some bats are eagles.

Conclusions:

1. Some frogs are not eagles.
2. No dog is a bat.
3. Some cats are bats.
4. No eagle is a cat.

Answer Options

Select any one option

Clear Answer

- Both conclusion 2 & 4 follow
- Conclusions 2, 3 & 4 follow
- Only conclusion 1 follows
- No conclusion follows

Submit test

‹ Previous

Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 28 min

Go to Analog ›

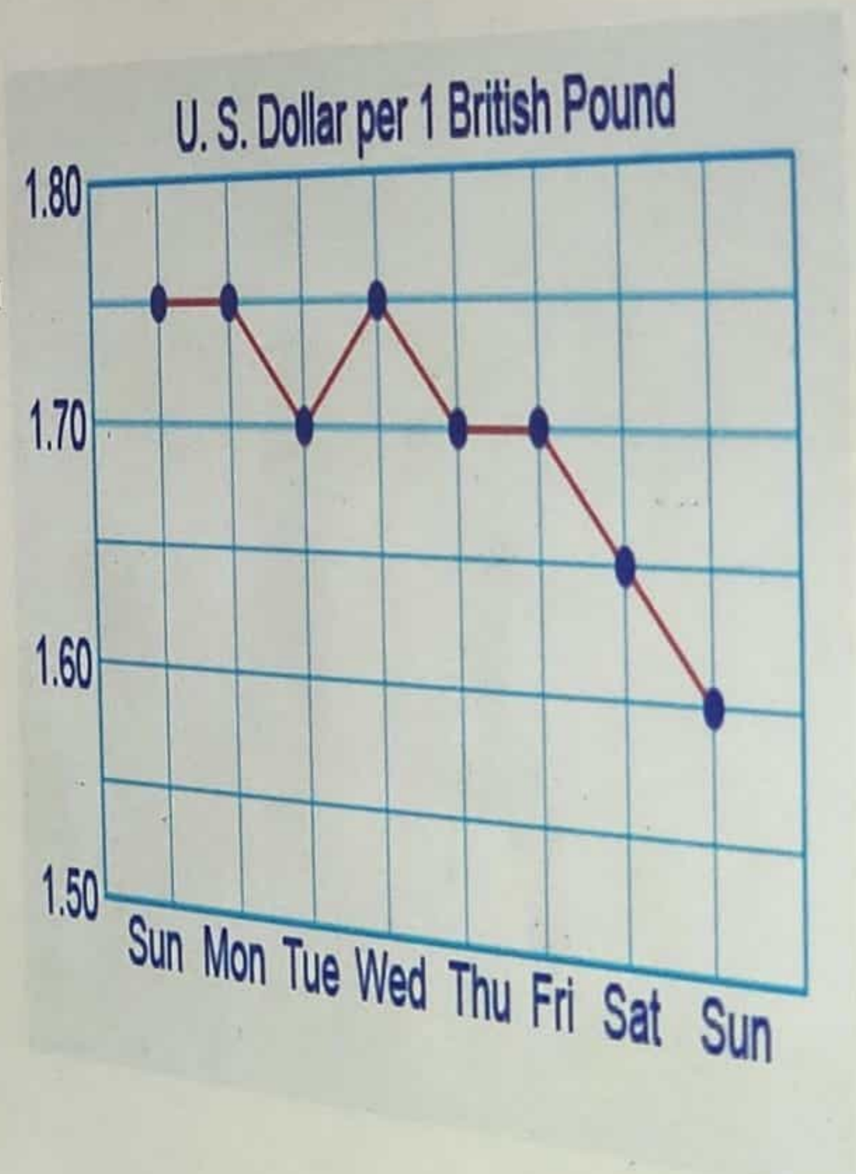
Digital

Submit test

14. The exchange rate dynamics

The given line graphs show the exchange rate dynamics during one week.

On the basis of the line graphs, how much did the dollar depreciate against the euro during this seven-day period?



Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 28 min

Go to Analog >

Digital

Submit test

13. Finding value of expression

In a certain system, two positive integers are subtracted in a certain way. Given below are some examples:

1. 62 - 56 = 0
2. 48 - 32 = -1
3. 55 - 31 = 4

Analyse the given expressions and determine the result of the expression **42 - 28**.

Answer Options

Select any one option

- 1
- 2
- 3
- 4

15. A chess club

A chess club has three teams **T1**, **T2** and **T3** out of which - one team always tells the truth, one team always lies and one alternates between telling the truth and lying. The teams get together only thrice a week, i.e., Tuesday, Friday and Sunday. One of these days, the three teams make the following statements:

T1: "Tuesday is the most comfortable day to get together and Friday is comfortable for T2".
T2: "Friday is comfortable for T3 and Sunday is comfortable for T1".
T3: "Tuesday is the most comfortable day to get together and T1 is comfortable with Sunday".

If all teams prefer a different day to get together, then which team is telling the truth?

Note: All teams works as a single union.

Answer Options

Select any one option

- T1
- T2
- T3
- Cannot be determined

Aptitude ends in 28 min

Aptitude

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Analog

Starts in 28 min

Go to Analog >

Digital

Submit test

14. The exchange rate

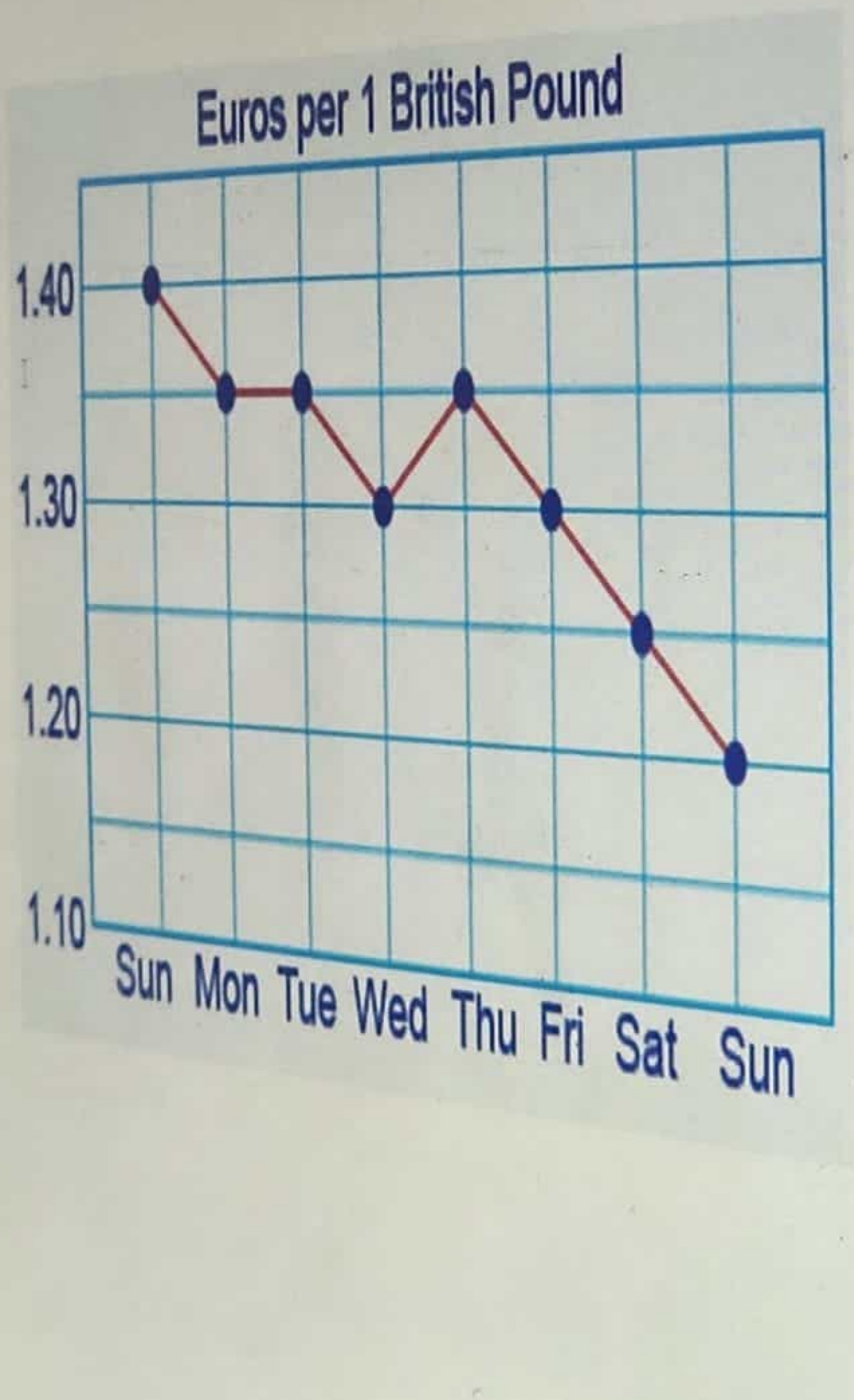
Answer Options

Select any one option

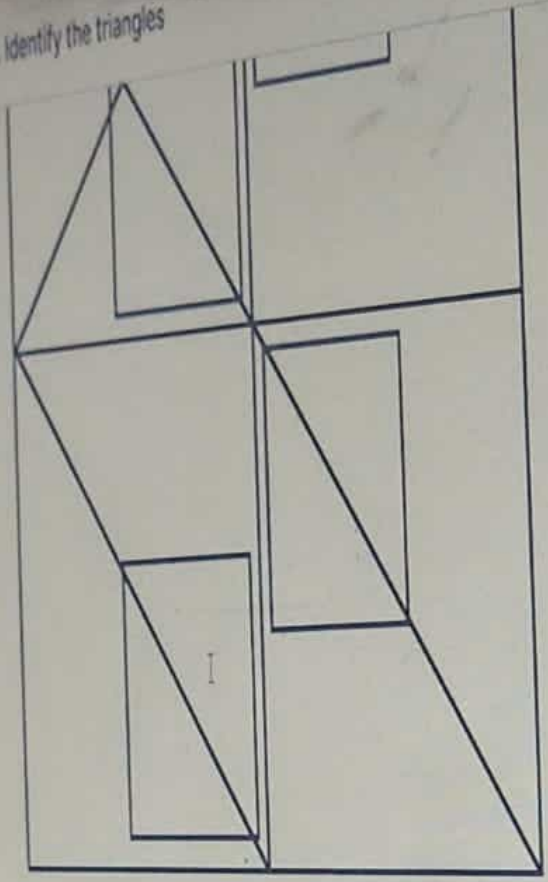
- 5.00%
- 5.50%
- 6.25%
- 7.50%

Clea

14. The exchange rate dynamics



17. Identify the triangles



Answer Options

Select any one option

21

20

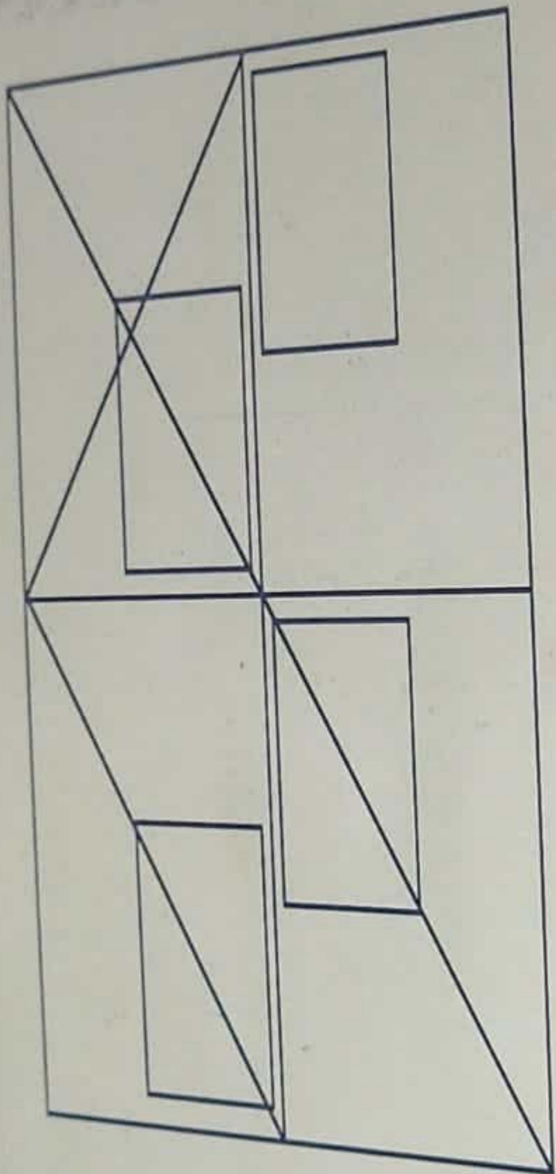
18

19

Previous

17. Identify the triangles

How many triangles can be found in the image given below?



Answer Options

Select any one option

21

20

16. A primary school teacher

A primary school teacher wants to arrange a visit to the museum which is 600 miles away from the school. She arranges a school bus for the same and ensures that the bus has a full tank. However, the bus driver does not remember the capacity of the gas tank nor does he remember the efficiency of the engine.

Which of these statements would be sufficient to determine if the bus has enough gas to reach the museum?

Statement 1: The driver is driving at a constant speed.
Statement 2: The driver realizes that he has three-quarters of the full tank of gas left after he drives 150 miles.
Statement 3: The driver realizes that he has three-quarters of the full tank of gas left after he drives 450 miles.

Answer Options

Select any one option

Clear Answer

- Both Statements 1 and 2 are sufficient
- Both Statements 1 and 3 are sufficient
- All three Statements are sufficient
- Statements 1 and 2 OR Statements 1 and 3 is sufficient
- None of the Statements are sufficient

1
2
3 → A
4 → D
5 → C
6 → D
7 → B
8 → B
9 → B [only 2 is incorrect]
10
11 → A
12 → only conclusion 1 follows
13 → C
14
15

16 → A
17 → A
~~8~~ 18 → ■.
~~8~~ 19 → D.
20. → A.

(Previous Next >

19. Selecting fudge balls

A dessert shop sells assorted boxes of chocolates every holiday season. Each of these boxes contains four caramel twists, three dark chocolate truffles and two fudge balls. John buys a box and successively picks three pieces of chocolates from it.

In the given context, what is the probability that John selects **exactly one fudge ball** from the box?

Answer Options

Select any one option

Clear Ans

- 1/2
- 2/3
- 3/4
- 1/4

...ule ends in 27 min

2 3
5 6
8 9
11 12
14 15
17 18
20

...ends in 27 min

to Analog >

s in 27

3

6

9

12

15

18

min

log ›

‹ Previous Next ›

18. A luxury perfume

A luxury perfume launched its perfumes on March 8, 1997 with a stock of x bottles of perfume. By Dec 31 of that same year, 2/7 of the x bottles are sold. This pattern continues such that by the end of each subsequent year 2/7 of the stock that was there at the beginning of that year is sold out.

Analyse the given scenario and determine the year in which the stock of these bottles is reduced to less than 1/4 of the original x bottles.

Note: *Assume that only one type of perfume is being sold and the bottles are not restocked.*

Answer Options

Select any one option

Clear Answer

- 2003
- 2002
- 2001
- 2000

Analog ends in 44 min

Aptitude

Submitted

Analog

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

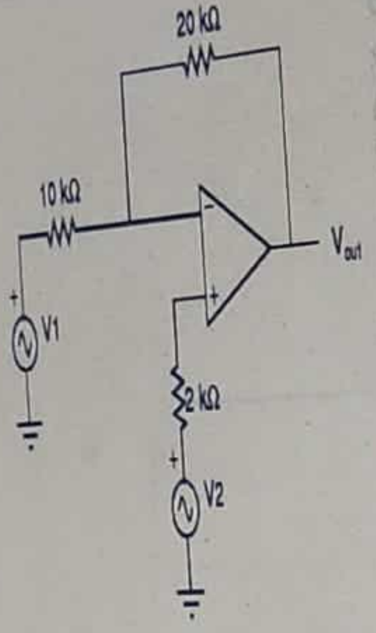
Digital

Starts in 44 min

Go to Digital ›

7. Output of the opamp

Consider the circuit shown in the figure. What is the output of the op amp?



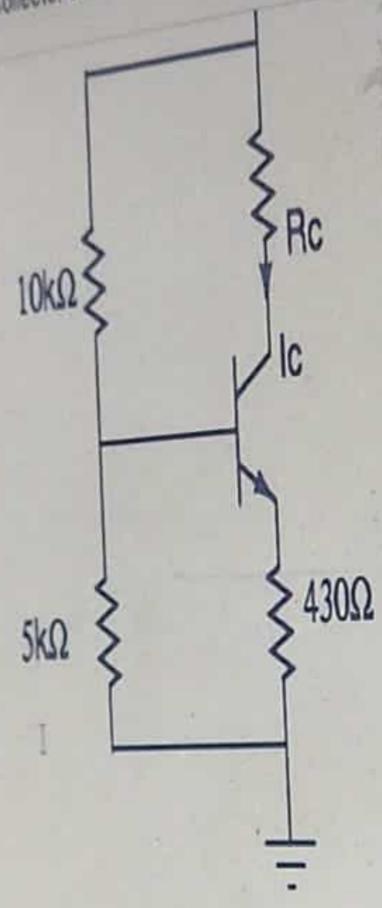
Answer Options

Select any one option

- 2 (v2 - v1)
- (3v2 - 2v1)
- (2v1 - 3v2)
- (3v1 - 2v2)

Submit test

11. Collector current of a transistor



Answer Options

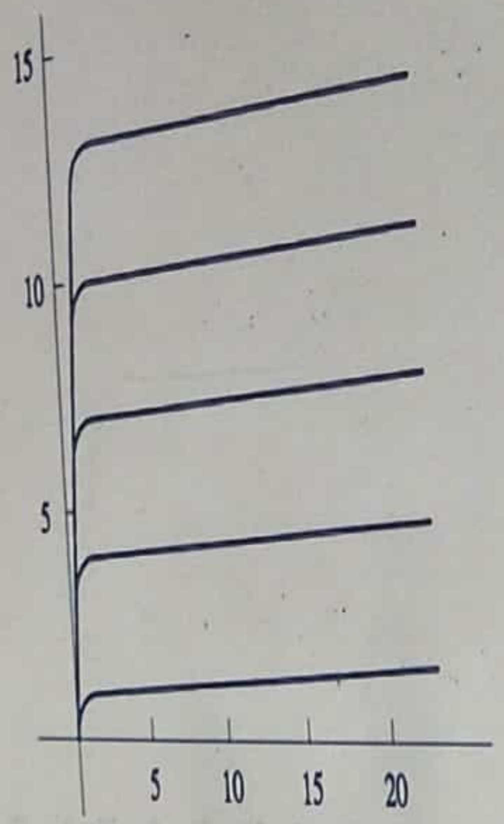
Select any one option

- 1 mA
- 5 mA
- 10 mA
- Can not be determined since Rc is not given.

alog ends in 43 min
ptitude
Submitted
Analog
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Digital
Starts in 43 min
Go to Digital ›

17. Characteristics of a BJT

Which of these choices is **incorrect** about the output characteristics of a BJT operating in common-emitter configuration?



Answer Options

Select any one option

- The graph has collector-emitter voltage along x-axis.
- The graph has collector current along the y-axis.
- Each curve in the graph corresponds to a different value of base current.
- None of these

s in 43 min

14. Using a 10 bit A/D converter

You are using a **10-bit** analog-to-digital converter with reference voltage of 5V to digitize an analog signal in the **0 to 5 V** range.

What is the maximum peak-to-peak ripple voltage that can be allowed in the reference voltage?

Answer Options

Clear

Select any one option

- nearly 100 mV
- nearly 50 mV
- nearly 25 mV
- nearly 5 mV

2 3 5 6 8 9 11 12 14 15 16 17 18 19 20

Digital

Starts in 43 min

Go to Digital ›

Submit test

‹ Previous

Next ›

Analog ends in 44 min

Aptitude

Submitted

Analog

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Digital

Starts in 44 min

Go to Digital ›

10. A series RLC circuit

A series **RLC** circuit has a resonance frequency of **40 MHz**.

Which of the given deductions can be made from the above information?

Answer Options

Select any one option

- At 40 MHz, the circuit shows a resistive impedance of R
- For frequencies lower than 40 MHz, the circuit behaves like an inductive circuit
- For frequencies greater than 40 MHz, the circuit behaves like a capactivite circuit
- All of the above

Analog ends in 43 min

Aptitude

Submitted

Analog

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

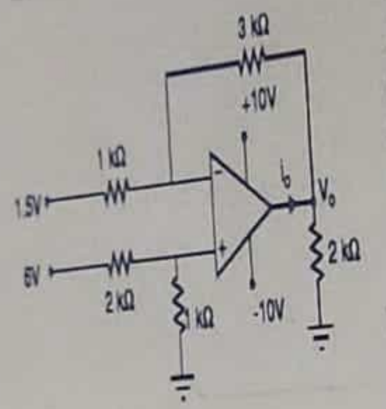
Starts in 43 min

Go to Digital >

< Previous

16. An ideal opamp circuit

Find the value of the output voltage V_o for the given ideal op amp circuit.



Answer Options

Select any one option

- 3.5V
- 4.0V
- 4.5V
- 3.0V

Analog ends in 43 min

Aptitude

Submitted

Analog

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 43 min

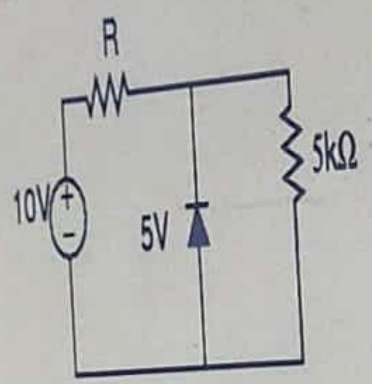
Go to Digital ›

Submit test

12. Voltage regulation

Note:

1. *RL denotes load resistance.*
2. The value of knee current $I_{knee} = 0$



Answer Options

Select any one option

- 4kΩ
- 5kΩ
- 3kΩ
- 2kΩ

Analog ends in 44 min

Aptitude

Submitted

Analog

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 44 min

Go to Digital ›

9. An ideal opamp

Which of the given statements is incorrect about an ideal op amp?

Answer Options

Select any one option

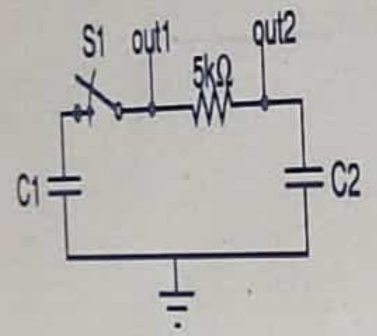
- Has infinite gain
- Has infinite input impedance
- Has infinite bandwidth
- Has infinite output impedance

5. Capacitors C1 and C2

You are working on the following passive circuit. The capacitor values are given as **C1 = 3 μF** and **C2 = 1 μF**. You have been informed that the capacitor **C1** is precharged to **10V**, while **C2** has no initial charge.

Also, the switch **S1** is initially open and is closed at time, **t= 0.**

What will be the final voltage obtained at the capacitors **C1 and C2?**



Answer Options

Select any one option

- C1 will reach 7.5V, C2 will reach 2.5V
- C1 will reach 2.5V, C2 will reach 7.5V
- C1 and C2 will both reach 7.5V
- C1 will reach 0, C2 will reach 10V

Clear Answ

Submit test

alog ends in 43 min

ptitude

Submitted

Analog

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Digital

Starts in 43 min

Go to Digital ›

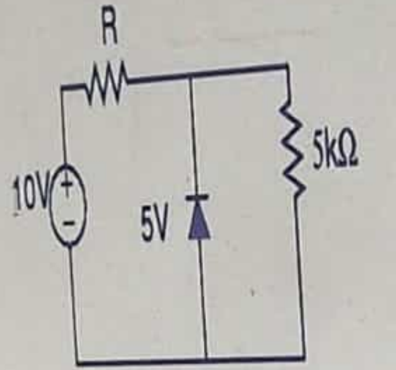
12. Voltage regulation

It is given that **Vin = 10V** and **RL = 5kΩ** for the given circuit. You now wish to use the same circuit as a regulator circuit.

What will be the maximum value of **R** such that a constant voltage is regulated across the load?

Note:

1. *RL denotes load resistance.*
2. The value of knee current $I_{knee} = 0$



Answer Options

Select any one option

- 4kΩ
- 5kΩ
- 3kΩ
- 2kΩ

Submit test

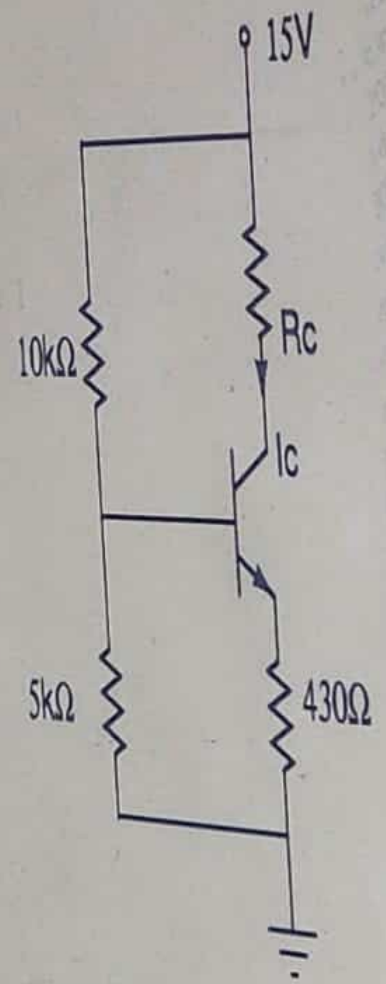
‹ Previous Next ›

11. Collector current of a transistor

You have designed the given circuit such that the transistor is operating in the active region. What will be the value of collector current Ic in the given scenario?

Note:

1. The current gain of the transistor, **β is very large**
2. The Base-emitter voltage of the amplifier is **0.7V.**



Answer Options

‹ Previous   Next ›

6. Modified output of the system

You have designed a linear time-invariant control system that produces an output **c(t)** for a certain input **r(t)**. You decide to modify **r(t)** by passing it through a block whose transfer function is $\mathbf{e^{-s}}$.

If the modified input is applied to the system, what will be the modified output of the system?

Note:

1. In the given transfer function $\mathbf{e^{-s}}$, **e** denotes an exponential function and **s** denotes the Laplace operator.

Answer Options

Clear Answ

Select any one option

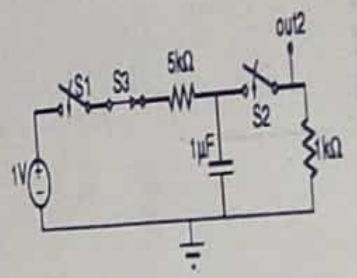
- c(t-1)
- c(t). u(t-1)
- c(t) / 1+e^t
- c(t) / 1+e^{-t}

19. A passive network

A passive network is connected, as shown in figure. It is observed that the switch **S1** is initially open, and is closed at time **t =0**. The switch S2 is also initially open and is closed at **3 ms**. However, the switch **S3** is initially closed and is opened only at time **2.5 ms**.

At what time will **out2** reach **0.2V**?

Note: Assume that capacitors have no initial charge.



Answer Options

Select any one option

Clear Answer

- t=2.67ms
- t=3.67ms
- t=4.67ms
- t=5.12ms

in 43 min

13. Series RLC circuit

You have a series RLC circuit with resistance **10 Ω**, inductance **1 mH**, and capacitance **10 μF**. Assume that a sinusoidal input of the form **V sin(ωt)** is applied to the circuit.

In the given context, for what value of **ω** will the current flowing in the circuit be in the form **I sin(ωt)**?

2 3 5 6 8 9 11 12 14 15 17 18 20

Answer Options

Clear An

Select any one option

- 10M rad/s
- 100M rad/s
- 100/(2*π) K rad/s
- 10 K rad/s

gital

Starts in 43 min

Go to Digital ›

Submit test

‹ Previous Next ›

Analog ends in 43 min

Aptitude

Submitted

Analog

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Digital

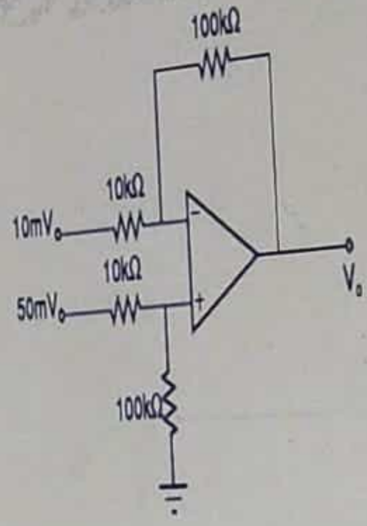
Starts in 43 min

Go to Digital ›

Submit test

20. An operational amplifier

In the circuit shown below, the operational amplifier is ideal in nature. Find the output voltage V_o.



Answer Options

Select any one option

- 500mV
- 100mV
- 600mV
- 400mV

15. Rms and average values

It is given that x_r and x_a are respectively the rms and average values of $x(t) = x(t-T)$. Similarly, y_r and y_a are respectively the rms and average values of $y(t) = kx(t)$.

Which of the following equations hold **TRUE** in the given scenario?

Note: *It is given that **k** and T are independent of **t**.*

Answer Options

Select any one option

- $y_a \neq kx_a\,;\, y_r = kx_r$
- $y_a = kx_a\,;\, y_r \neq kx_r$
- $y_a \neq kx_a\,;\, y_r \neq kx_r$
- $y_a = kx_a\,;\, y_r = kx_r$

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2 3 5 6 8 9 10 11 12 13 14 15 16 17 18 19 20
Digital
Starts in 43 min
Go to Digital ›
Submit test

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Analog ends in 43 min

Aptitude

Submitted

Analog

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

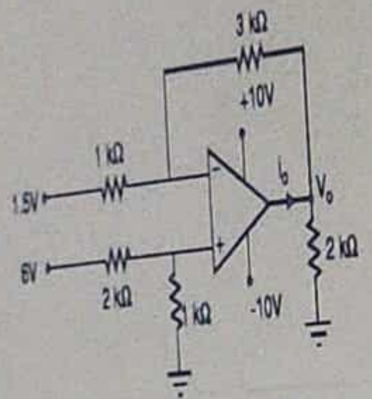
Digital

Starts in 43 min

Go to Digital >

16. An ideal opamp circuit

Find the value of the output voltage V_o for the given ideal op amp circuit.



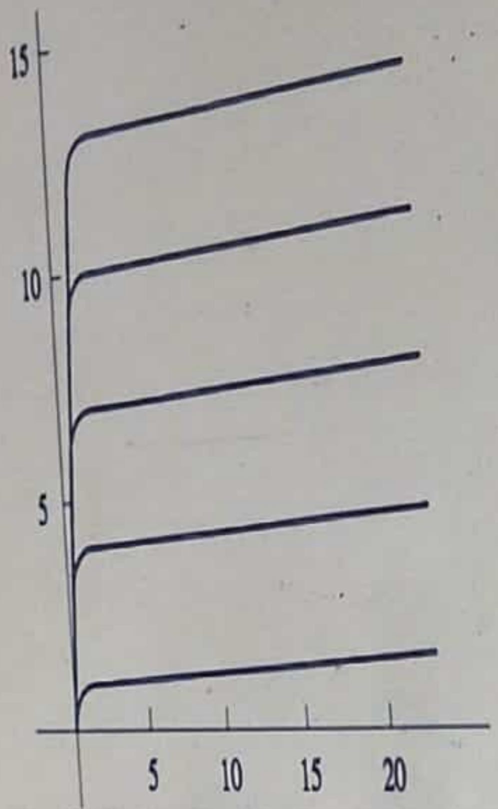
Answer Options

Select any one option

- 3.5V
- 4.0V
- 4.5V
- 3.0V

Submit test

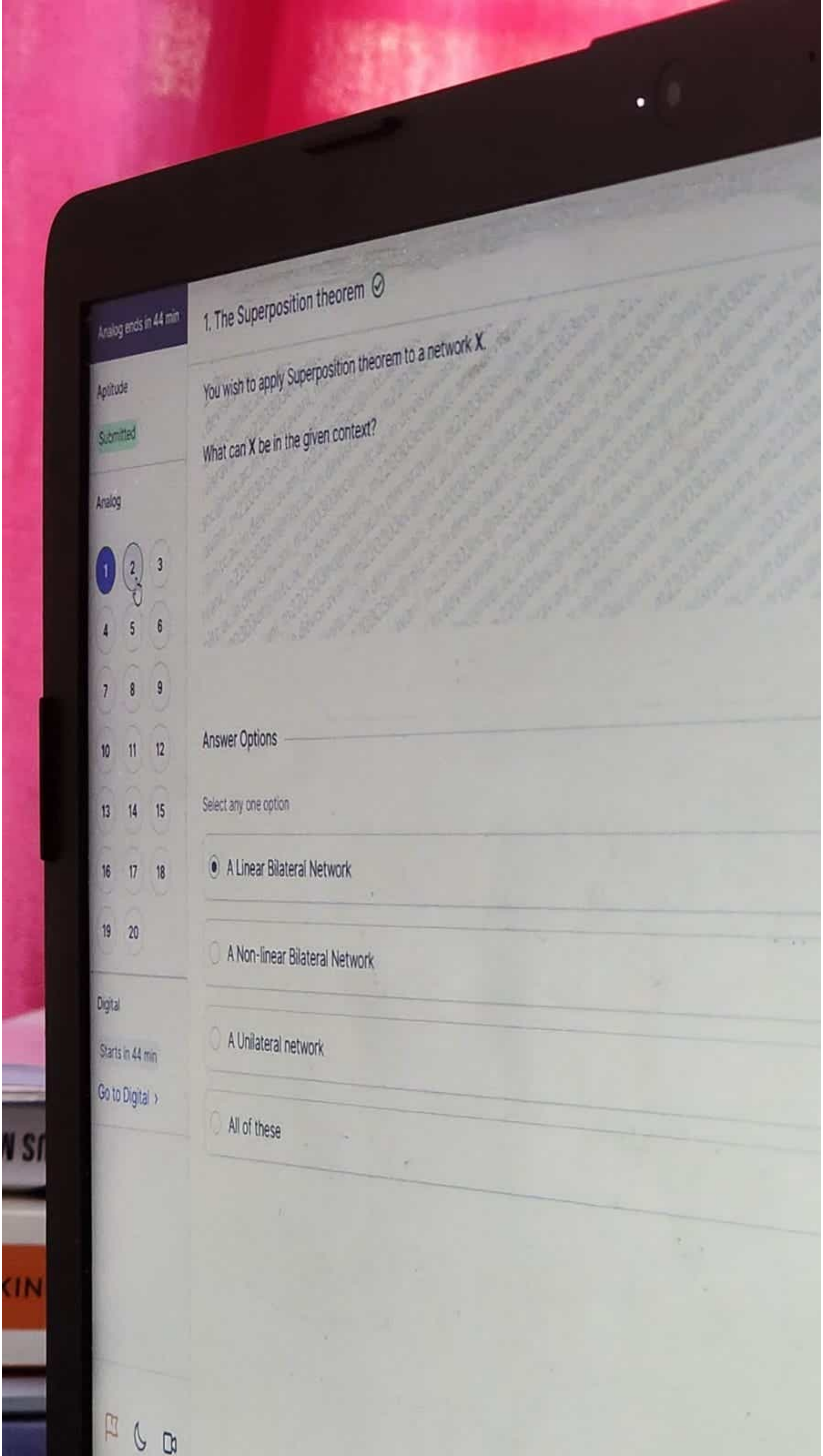
17. Characteristics of a BJT



Answer Options

Select any one option

- The graph has collector-emitter voltage along x-axis.
- The graph has collector current along the y-axis.
- Each curve in the graph corresponds to a different value of base current.
- None of these



Analog ends in 44 min
1. The Superposition theorem
You wish to apply Superposition theorem to a network X.
What can X be in the given context?
Aptitude
Submitted
Analog
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20
Answer Options
Select any one option
A Linear Bilateral Network
A Non-linear Bilateral Network
A Unilateral network
All of these
Digital
Starts in 44 min
Go to Digital ›

s in 44 min

2. Maximum power

A voltage source $v(t) = V*\sin 450*\pi t$ Volts having an internal impedance of $(5+j12)\Omega$ is connected to a purely resistive load R_L

What should be the value of R_L (in Ω) such that it can draw the maximum power out of the source?

< Previous

3
6
9
12
15
18

Answer Options

Select any one option

- 2Ω
- 3Ω
- 4Ω
- 13Ω

4 min

ital ›

Analog ends in 44 min

Aptitude

Submitted

Analog

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 44 min

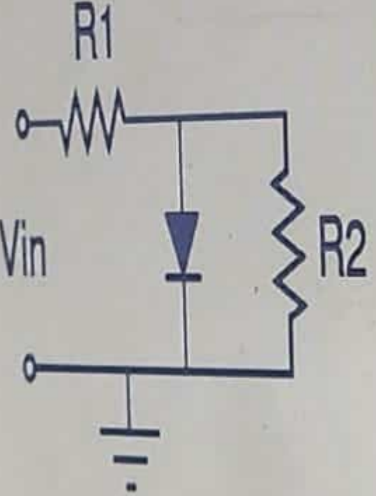
Go to Digital ›

4. Minimum value of resistance

The given diode circuit has the following parameters:

1. Input supply voltage, **Vin**=4V
2. Resistance **R1** = 6.6kΩ
3. Cut-in voltage of diode, **Vγ** = 0.7V

Calculate the minimum value of resistance **R2** for which the diode will be **ON.**



Answer Options

Select any one option

- 1.4 kΩ
- 1.8 kΩ
- 2.4 kΩ
- 3 kΩ

Analog ends in 44 min

Aptitude

Submitted

Analog

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

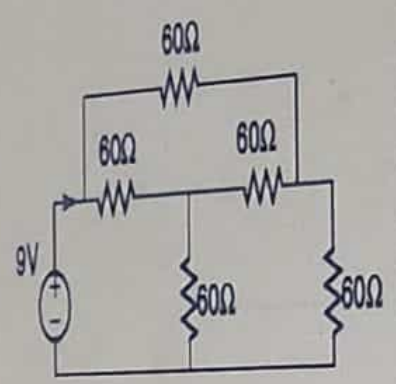
Starts in 44 min

Go to Digital ›

3. The input resistance

What is the input resistance as seen from the source side in the given circuit?

Hint: *Transform the T-network to a Delta-network.*



Answer Options

Select any one option

- 270 ohm
- 180 ohm
- 120 ohm
- 60 ohm

Analog ends in 44 min

Aptitude

Submitted

Analog

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 44 min

Go to Digital ›

8. Resolution of an ADC

You have a **10-bit ADC** with input ranging from **0 to 10V**.

What will be the resolution **(step-size)** of the ADC?

Answer Options

Select any one option

- 0.977 mV
- 9.77 mV
- 97.7 mV
- 977 mV

Analog ends in 43 min

Aptitude

Submitted

Analog

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Digital

Starts in 43 min

Go to Digital ›

Submit test

‹ Previous Next

18. A cascade of three linear, time-invariant system

It is given that a cascade of three linear, time-invariant systems is causal and unstable.

Which of these conclusions can be made about the cascaded system?

Answer Options

Select any one option

- Each system in the cascade is individually causal and unstable.
- At least one system is unstable and at least one system is causal.
- At least one system is causal and all systems are unstable.
- The majority of the systems are unstable and causal.

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

‹ Previous Next ›

4. A bistable multivibrator

You want to design a bistable multivibrator to increase the frequency at which the data is stored. When doing so, you want to ensure that the data is stored at the rising and falling edge of the clock. However, due to lack of resources, you only have MUXes to implement this.

How many MUXes will you need to achieve your objective?

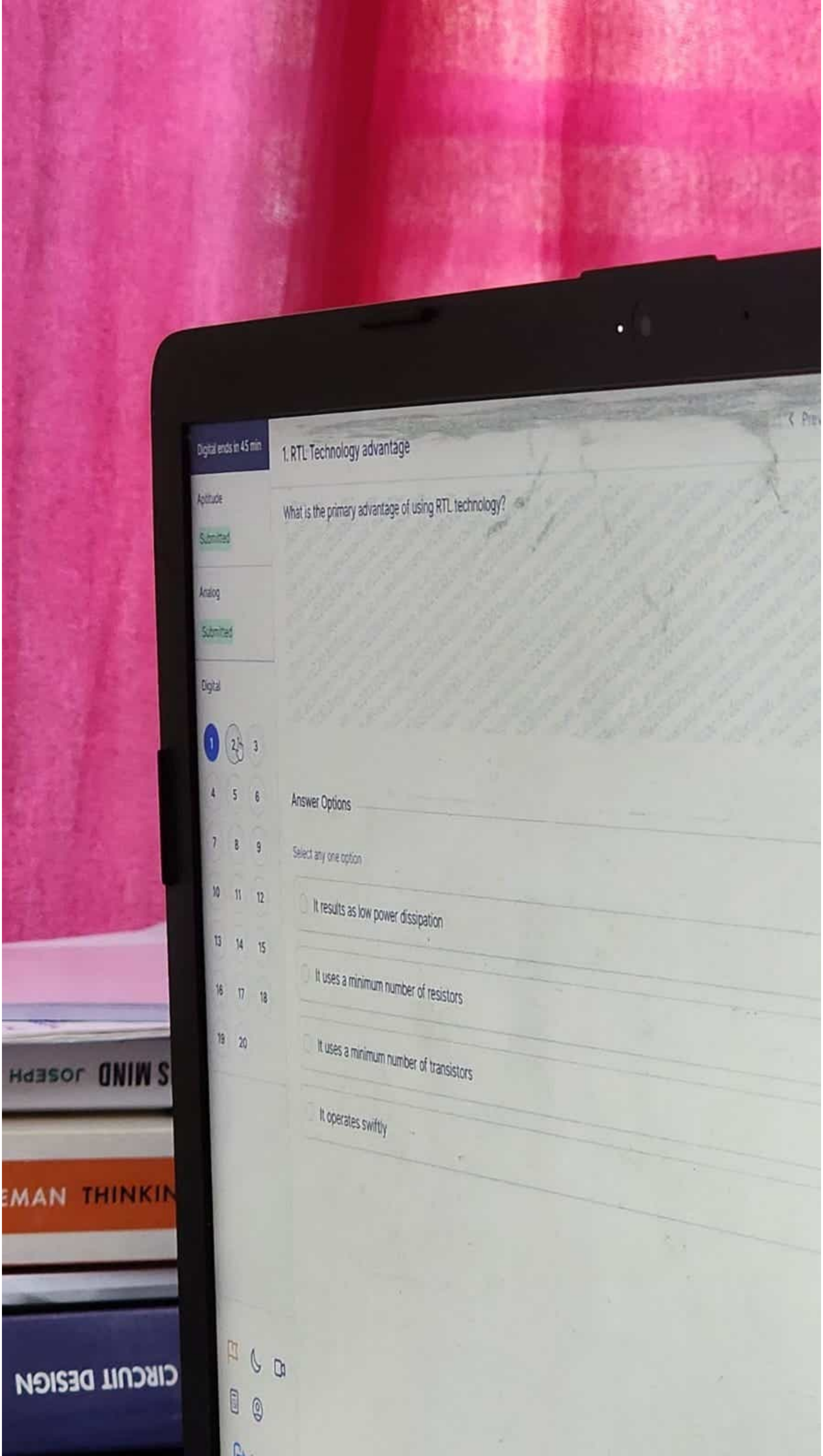
Answer Options

Select any one option

Clear Answer

- 2
- 3
- 4
- 5

Submit test


Digital ends in 45 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

1. RTL Technology advantage

What is the primary advantage of using RTL technology?

Answer Options

Select any one option

- It results as low power dissipation
- It uses a minimum number of resistors
- It uses a minimum number of transistors
- It operates swiftly

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

❮ Previous Next ❯

6. The equivalent function

Which of these functions is the same as the following function?

$f = ((x.y') + (x'.y))'$

Answer Options

Select any one option

Clear Answer

- $f = (x+y)(x'+y')$
- $f = ((x'+y')(x+y))'$
- $f = (x.y)'(x'.y')$
- None of these

Submit test

Digital ends in 44 min

7. Code synthesis

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

```
wire [255 : 0] a;
wire [255 : 0] b;
wire [255 : 0] c;

assign c = sel_line ? a : b ;
```

Which of the following can be inferred from the above given RTL code snippet?

1. The code on synthesis will translate into a relatively slow design.
2. The sel_line has 256 loads.
3. The code synthesizes into 256 parallel MUX structures.

Answer Options

Select any one option

- 1,2
- 1,3
- 2,3
- 1,2,3

Clear Answer

‹ Previous Next ›

Submit test

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Submit test

‹ Previous Next ›

8. Branches in a graph

The graph of a network has 10 nodes and 4 independant loops.

Find the number of branches in the graph.

Answer Options

Select any one option

Clear Answer

- 11
- 13
- 14
- 12

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Submit test

< Previous Next >

10. Executing a RISC instruction set

You are executing an instruction set in a RISC processor.

In how many instruction stages can you complete this execution?

Answer Options

Select any one option

Clear Answer

- 4
- 5
- 3
- 7

Digital ends in 45 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

Submit test

‹ Previous Next ›

3. A flash memory

Which of these statements doesn't describe Flash memory?

Answer Options

Select any one option

Clear Answ

- It can erase data byte-wise.
- It is used to rewrite the data frequently.
- NOR gates are used to design this memory.
- Both Choice 1 and 3
- None of these

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Submit test

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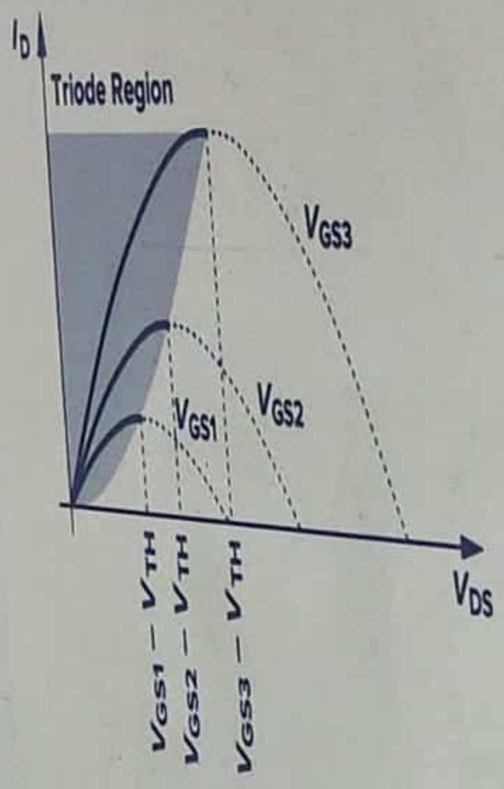
11. The IV characteristics of an nMOS

The IV characteristics of an nMOS, as predicted by the standard equation of drain current ID, are given in the following graph.

In reality, the drain current does not follow the parabolic behavior and ID becomes relatively constant, and we say the device operates in the "saturation" region. What could be a valid reason behind this?



Answer Options

Select any one option

- Some channel still exists even when VDS> VGS - VTH
- Channel is pinched off but current flows due to minority charge careers
- Upon passing the pinch-off point, the electrons simply shoot through the depletion region near the drain junction ... rain terminal.

Clear Answer

...al ends in 45 min

...titude

...bmitted

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Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

‹ Previous | Next ›

2. No. of cache sets

You are working on a 16-bit microprocessor that uses a 2-way set-associative cache with a total capacity of 256 bytes. It is given that each cache line contains 8 bytes.

How many cache sets are there in the cache?

Answer Options

Select any one option

Clear Answer

- 8
- 16
- 32
- 64

Submit test

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3

4 5 6

7 8 9

10 11 12

13 14 15

16 17 18

19 20

‹ Previous Next ›

9. Conditioning in Latch

Assume that you have designed a latch using NAND and NOR Gates.

Which condition enables this configuration to remain in the latched condition?

Answer Options

Select any one option

Clear Answer

- Low input voltages
- Synchronous operation
- Gate impedance
- Cross-Coupling

Submit test

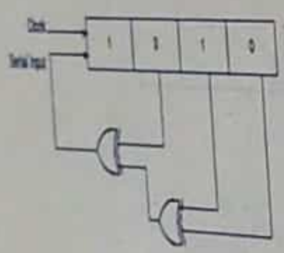
< Previous Next >

5. Working on a shift register

You are working on the given shift register. You have initially loaded it with the bit pattern **1010**. Subsequently, the shift register is clocked.

On clocking, it is observed that with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB).

How many clock pulses will the shift register take such that its content becomes **1010** again?



Answer Options

Select any one option

3

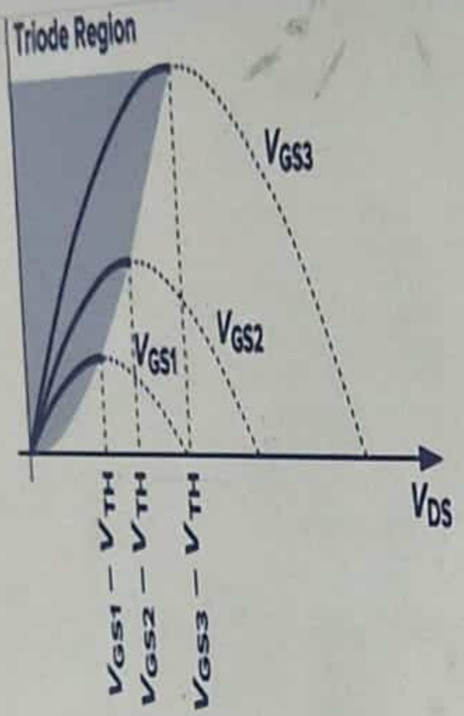
7

11

15

Clear Answer

11. The IV characteristics of an nMOS



Answer Options

Select any one option

Some channel still exists even when VDS> VGS - VTH

Channel is pinched off but current flows due to minority charge careers

Upon passing the pinch-off point, the electrons simply shoot through the depletion region near the drain junction and arrive at the drain terminal.

VDS is so high that it causes the current flow from the bulk to the drain terminal via the substrate.

Clear Answer

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Submit test

‹ Previous Next ›

12. Response of a system

A control system is defined by the given mathematical relationship:

> d^2x/dt^2 + 6. dx/dt+ 5x = 12(1-e^-2t), where x(t=0) = 0 and x'(t=0) = 0

Calculate the response of the system as $t \to \infty$.

Answer Options

Select any one option

Clear Answer

- x = 6
- x = 2
- x = 2.4
- x = -2

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

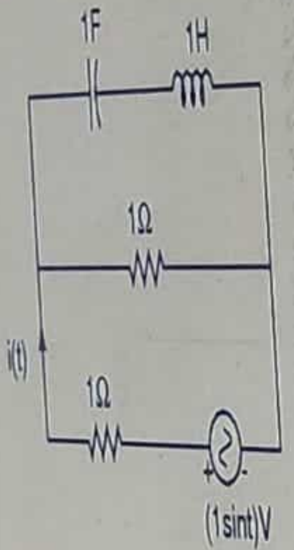
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Submit test

‹ Previous | Next ›

13. Finding rms value of current

Find the rms value of the current i(t) in the circuit shown below.



Answer Options

Select any one option

- 1/2 A
- 1/√2 A
- 1A
- √2 A

Clear Answer

Digital ends in 44 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

‹ Previous   Next ›

14. Reducing dynamic power consumption

You want to reduce the dynamic power consumption in a certain system. It is given that the system comprises of a newly developed block which will multiply the data passed into the system at every clock cycle. The steady state of the system is close to 0. However, due to noises, it is expected that the system will be fluctuating about the steady state.

You want to ensure that the dynamic power loss is minimum as the value fluctuates. In which of these forms should numbers be stored in the system so as to achieve the same?

Answer Options

Select any one option

Clear Answer

- 1s Complement
- 2s Complement
- Signed Magnitude
- Dynamic Power is independent of Number System Representation

Submit test

Digital ends in 44 min

15. Using sizeof()

Predict the output of the following code snippet.

```
class temp{
int x;
temp node;
float y;
};
main()
{
temp head = new temp();
display(sizeof(head), sizeof(temp));
}
```

Answer Options

Select any one option

- 8 24
- 24 24
- 4 12
- 8 12

Clear Answer

‹ Previous   Next ›

Aptitude — Submitted
Analog — Submitted
Digital

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

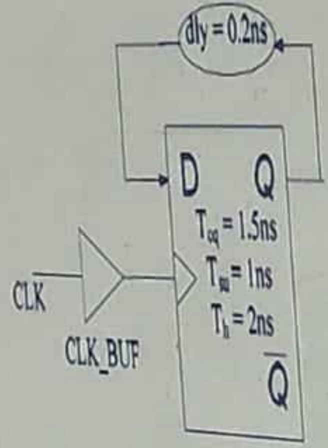
Submit test

16. Type of violation and delay

Analyze the given circuit and choose the option that answers the below question correctly.

A. What is the type of violation of the circuit?
B. What is the delay of CLK_BUF that will increase the frequency of operation of the circuit by 1.5x?

Note: Assume delay of CLK_BUF is 0.



Answer Options

Select any one option

A - Setup violation
B - 0.5ns

Clear Answer

A - Hold violation
B - CLK_BUF delay does not affect frequency of operation

A - Hold Violation
B - 0.5 ns

A - No violation
B - CLK_BUF delay does not affect frequency of operation

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

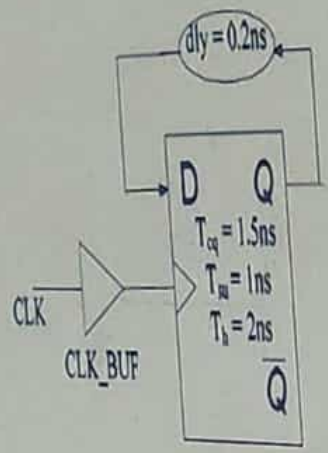
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

‹ Previous   Next ›

16. Type of violation and delay

b. What is the delay of CLK_BUF that will increase the frequency of operation of the circuit by 1.5x?

Note: Assume delay of CLK_BUF is 0.



Answer Options

Select any one option

Clear Answer

- A - Setup violation
  B - 0.5ns
- A - Hold violation
  B - CLK_BUF delay does not affect frequency of operation
- A - Hold Violation
  B - 0.5 ns
- A - No violation
  B - CLK_BUF delay does not affect frequency of operation

Submit test

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

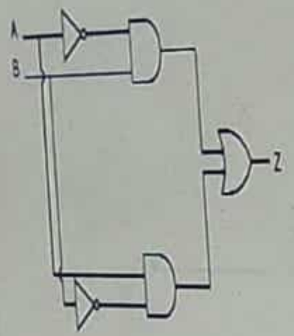
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

‹ Previous Next ›

17. The correct equivalent circuit

In the circuit shown below, **A** and **B** are the digital inputs and **Y** is the digital output.

Identify the correct equivalent circuit for the same.



Answer Options

Select any one option

- XOR gate
- XNOR gate
- NAND gate
- NOR gate

Clear

Submit test

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Submit test

‹ Previous | Next ›

18. Understanding strings in C

Predict the output of the following C code snippet.

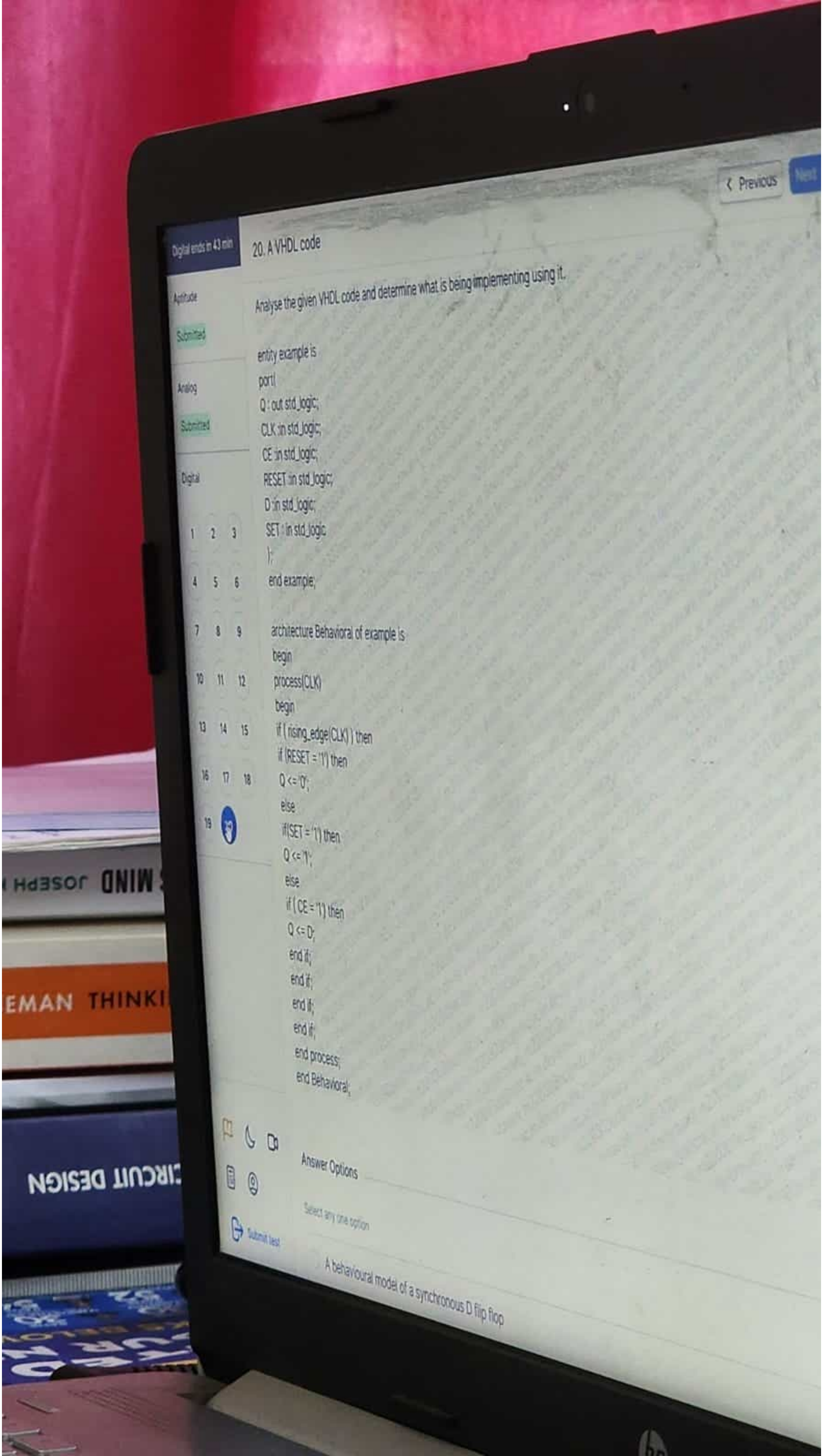
```
#include <stdio.h>
int main() {
  char ABC[5] = "hello";
  printf("%d", *(ABC + 4));
  return 0;
}
```

Answer Options

Select any one option

Clear Ans

- 111
- 0
- Garbage value
- Runtime error


Digital ends in 43 min
20. A VHDL code
Previous
Next
Aptitude
Submitted
Analog
Submitted
Digital
1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19
Analyse the given VHDL code and determine what is being implementing using it.
entity example is
port(
Q : out std_logic;
CLK :in std_logic;
CE :in std_logic;
RESET :in std_logic;
D :in std_logic;
SET : in std_logic
);
end example;
architecture Behavioral of example is
begin
process(CLK)
begin
if (rising_edge(CLK)) then
if (RESET = '1') then
Q <= '0';
else
if(SET = '1') then
Q <= '1';
else
if (CE = '1') then
Q <= D;
end if;
end if;
end if;
end if;
end process;
end Behavioral;
Answer Options
Select any one option
A behavioural model of a synchronous D flip flop
Submit test
JOSEPH
MIND
EMAN THINKI
CIRCUIT DESIGN

Digital ends in 43 min

Aptitude

Submitted

Analog

Submitted

Digital

1 2 3
4 5 6
7 8 9
10 11 12
13 14 15
16 17 18
19 20

Submit test

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19. Reading values from a DRAM Memory

The following steps need to be carried out while reading values from a DRAM Memory.

1. Precharging the column line parasitic capacitance to half-threshold.
2. Asserting a single row line.
3. Changing of the bit line voltage slightly above or below threshold depending on whether the value stored is 0 or 1.
4. Reading and amplification of the signal on the column line.

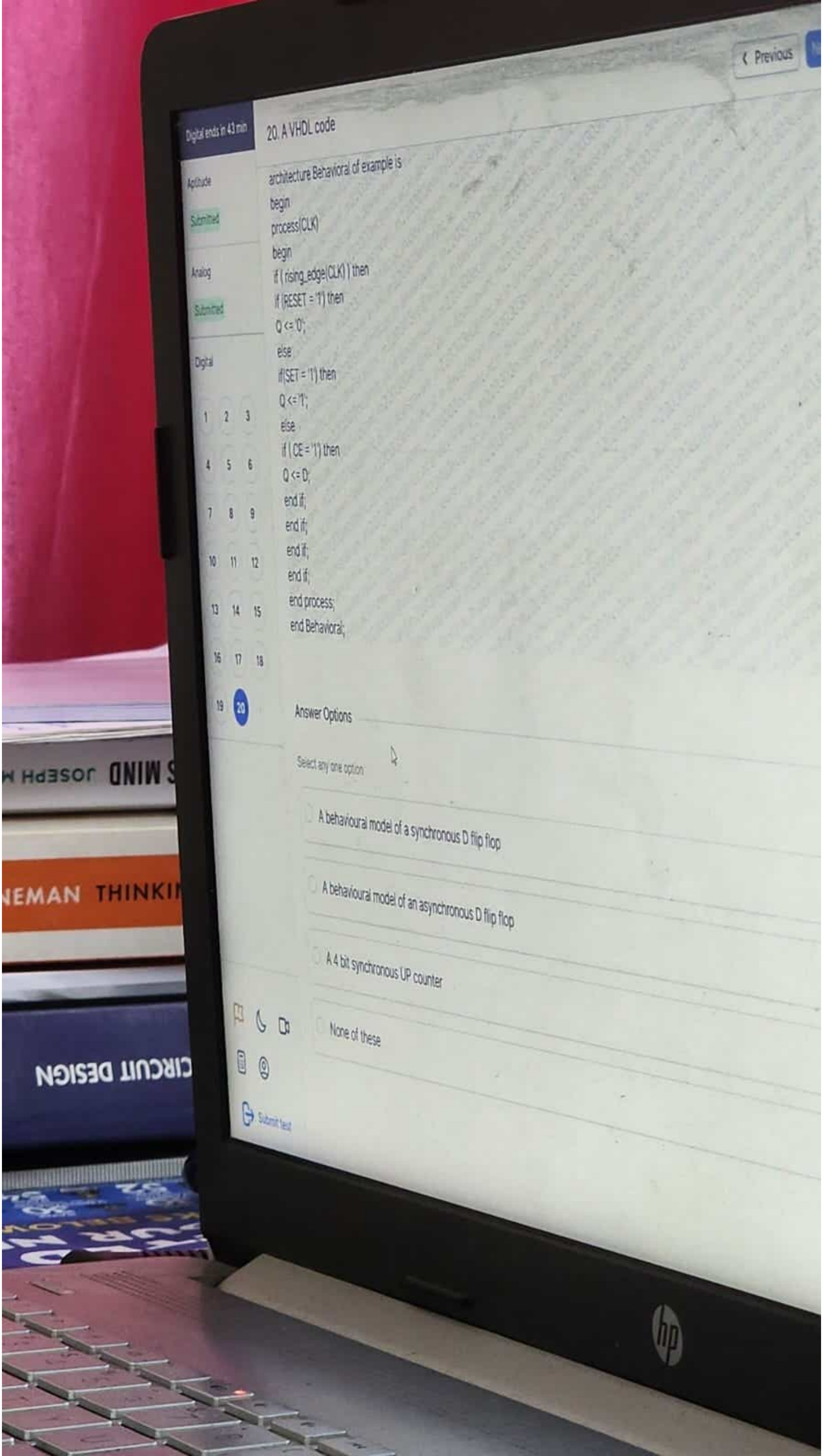
Which of these steps will be impacted directly if the storage capacitor used in a DRAM cell is small?

Answer Options

Select any one option

Clear Answer

- 1
- 2
- 3
- 4



Previous
Digital ends in 43 min
Aptitude
Submitted
Analog
Submitted
Digital
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
20. A VHDL code
architecture Behavioral of example is
begin
process(CLK)
begin
if (rising_edge(CLK)) then
if (RESET = '1') then
Q <= '0';
else
if(SET = '1') then
Q <= '1';
else
if (CE = '1') then
Q <= D;
end if;
end if;
end if;
end if;
end process;
end Behavioral;
Answer Options
Select any one option
A behavioural model of a synchronous D flip flop
A behavioural model of an asynchronous D flip flop
A 4 bit synchronous UP counter
None of these
Submit test

Technical

1 → A
2 → D
3 → D
4 → A.
5
6 → A
7 → B
8 → B
9 → D.
10 → D/A

11 → C.
12 → B.
13 → D
14 → D (Nearly 5mV)
15 → A
16 → A
17 → D [None of these]
18 → A
19
20 → D [400mV]